Medicaid Expenditure Projection Report

Project scope:

Forecast Medicaid expenditures for Michigan and the United States using annual data from 2013 to 2025.

Generate a 10-year forward projection for 2026 to 2035 with 95 percent confidence intervals.

Key results:

Michigan best model: prophet (holdout MAPE 6.71%)

Michigan projected spending: 28.87*Bin*2026*to*44.25B in 2035

United States best model: prophet (holdout MAPE 0.64%)

United States projected spending: 1099.60*Bin*2026*to*1736.05B in 2035

Included in this PDF:

Requirement checklist, confidence interval tables, model comparison summary, and all key visualization figures.

Output & Visualization Requirement Checklist

Requirement	Status	Evidence
10-year projection table with 95 percer	Completed	Included in report tables
Historical vs. projected expenditure plo	Completed	Included in report figures
HTML export	Completed	index.html in packaged folder
PDF export	Completed	This PDF file
Optional dashboard	Not required	Not included

Michigan 2026-2035 Forecast With 95 Percent Confidence Intervals

Geography	Year	Forecast (Billions)	Lower 95% Bound	Upper 95% Bound
Michigan	2026	28.87	28.06	29.65
Michigan	2027	30.58	28.65	32.38
Michigan	2028	32.29	28.79	35.86
Michigan	2029	34.00	29.15	39.62
Michigan	2030	35.71	29.04	43.28
Michigan	2031	37.42	28.99	47.58
Michigan	2032	39.13	28.79	51.40
Michigan	2033	40.84	28.29	55.67
Michigan	2034	42.54	27.82	60.07
Michigan	2035	44.25	26.15	65.00

United States 2026-2035 Forecast With 95 Percent Confidence Intervals

Geography	Year	Forecast (Billions)	Lower 95% Bound	Upper 95% Bound
United States	2026	1099.60	1074.93	1125.31
United States	2027	1170.28	1117.23	1236.57
United States	2028	1241.14	1151.21	1355.00
United States	2029	1311.82	1166.59	1472.89
United States	2030	1382.49	1185.15	1591.69
United States	2031	1453.17	1198.28	1718.69
United States	2032	1524.03	1210.56	1850.06
United States	2033	1594.71	1221.95	1977.00
United States	2034	1665.38	1211.96	2125.83
United States	2035	1736.05	1194.91	2298.22

Model Comparison Summary

Geography	Model Family	Model	MAE	RMSE	MAPE
Michigan	Baseline	linear_regression	1,848,128,323.06	1,983,664,301.41	7.12%
Michigan	Baseline	polynomial_regression_d	2,432,728,381.78	2,593,707,374.40	9.38%
United States	Baseline	linear_regression	112,706,434,239.52	116,157,169,999.68	11.64%
United States	Baseline	polynomial_regression_d	47,102,709,358.58	47,978,768,922.10	4.89%
Michigan	Time Series	arima	1,874,605,435.81	2,043,074,937.14	7.20%
Michigan	Time Series	holt_winters	2,026,125,304.66	2,165,374,071.43	7.81%
Michigan	Time Series	prophet	1,742,600,946.76	1,867,808,861.48	6.71%
United States	Time Series	arima	33,126,607,683.76	35,702,415,248.31	3.40%
United States	Time Series	holt_winters	26,688,505,965.20	29,144,349,989.41	2.74%
United States	Time Series	prophet	6,119,587,188.18	8,110,079,553.23	0.64%

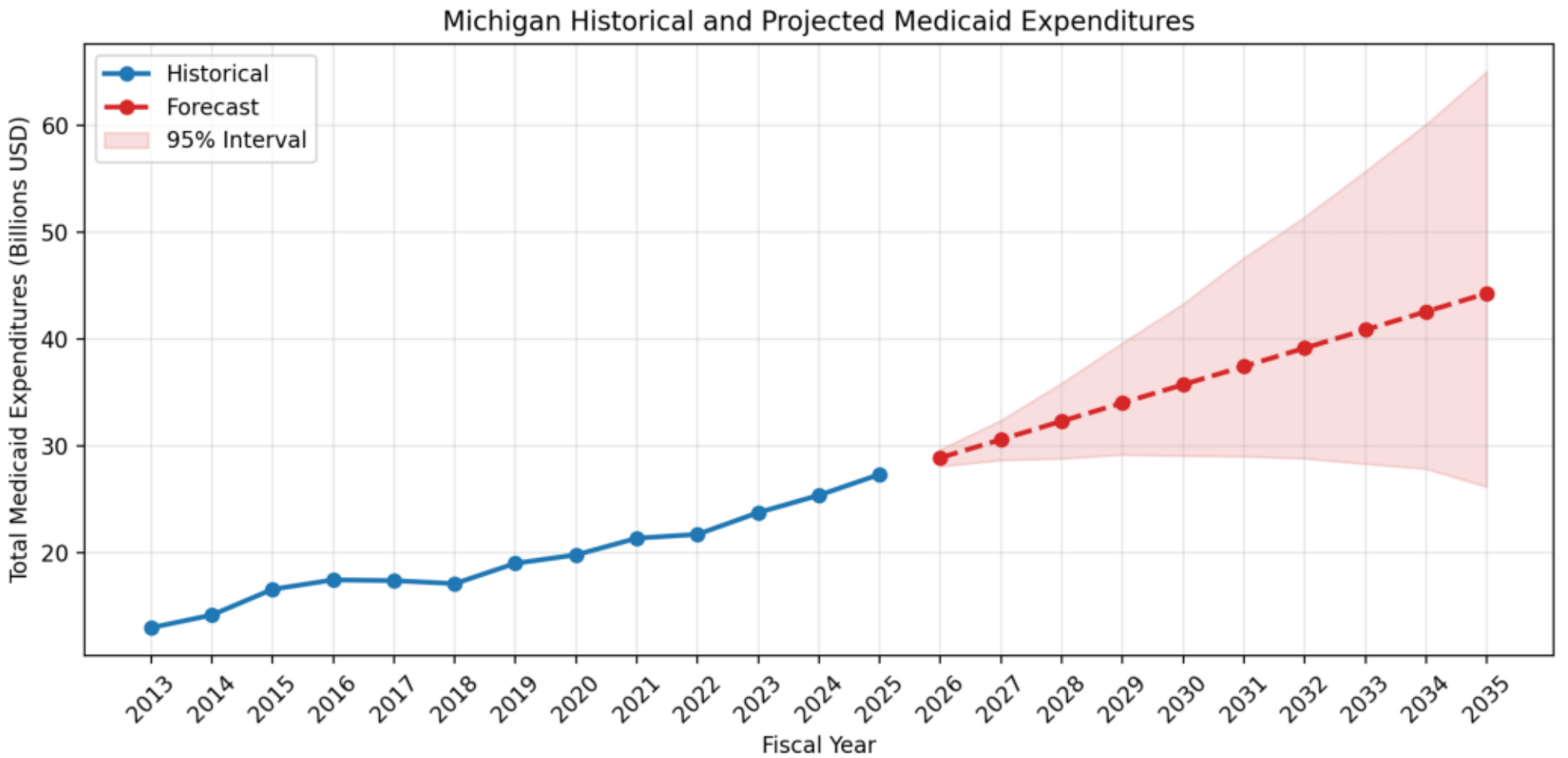
Final Historical vs. Projected Expenditure Plots

The following pages contain the figures generated for this section.

Each figure is reproduced directly from the project outputs folder.

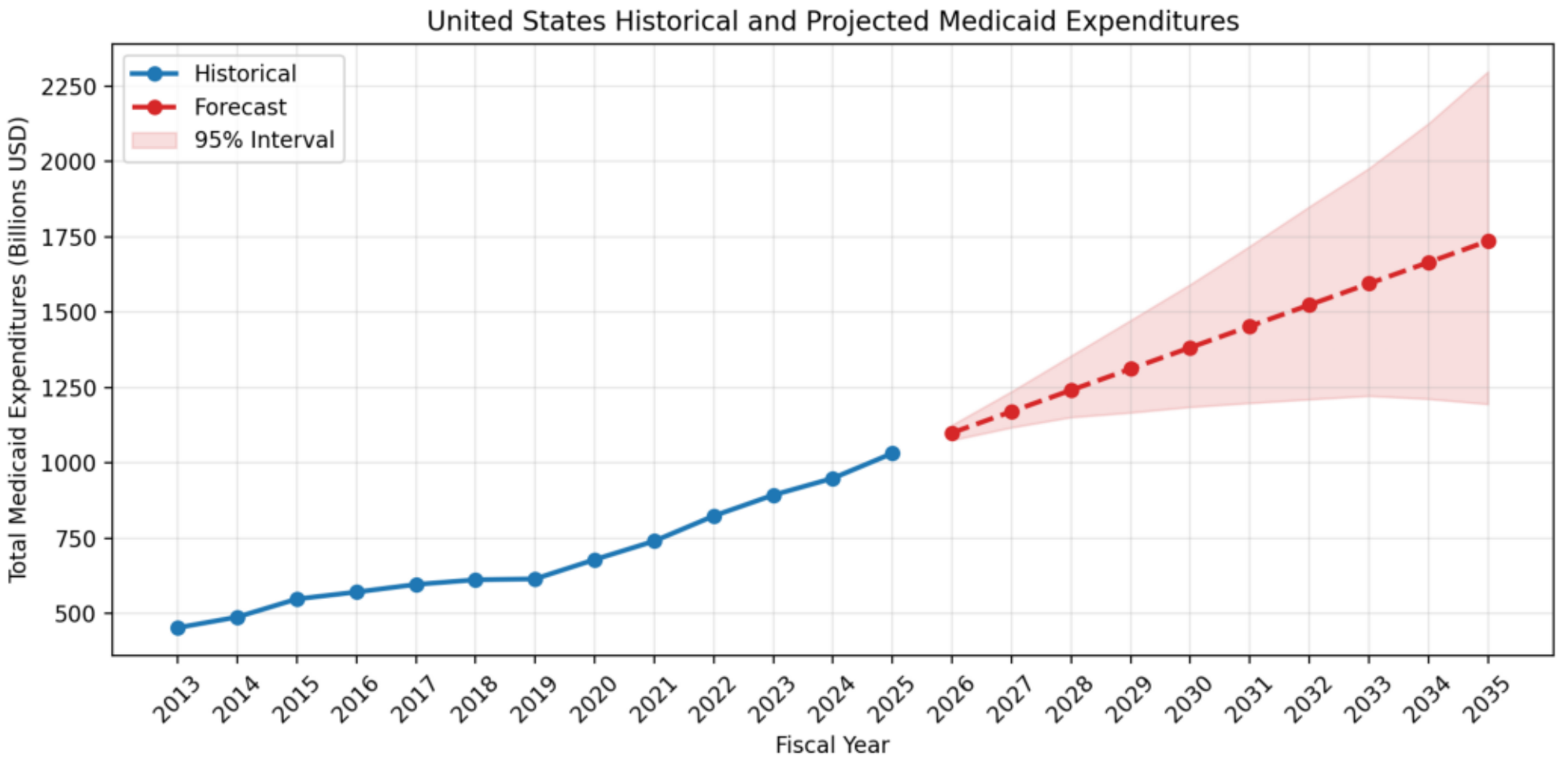
Michigan Final Projection

Historical values plus 2026-2035 forecast with 95 percent interval.



United States Final Projection

Historical values plus 2026-2035 forecast with 95 percent interval.



EDA Visualizations

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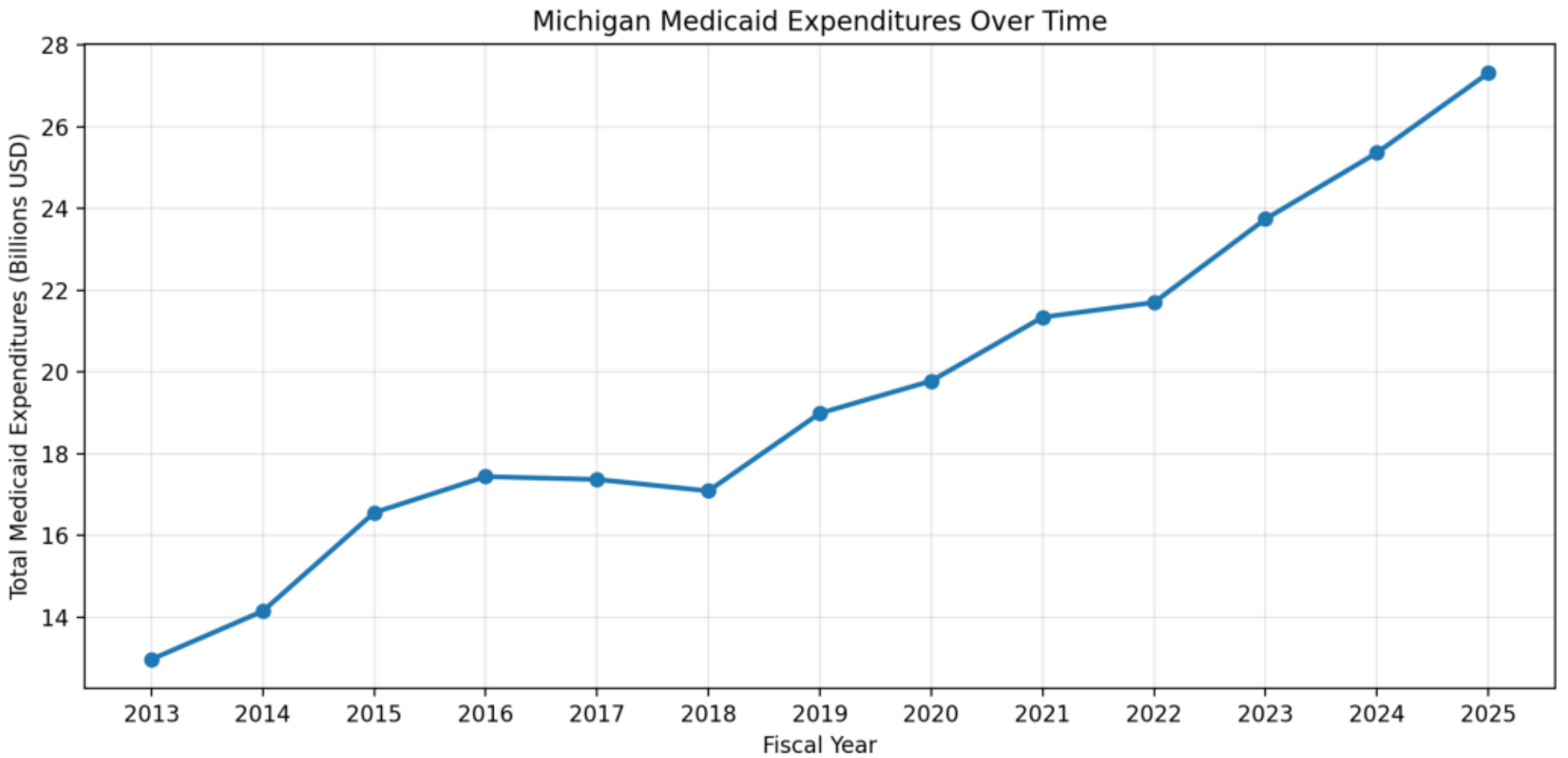
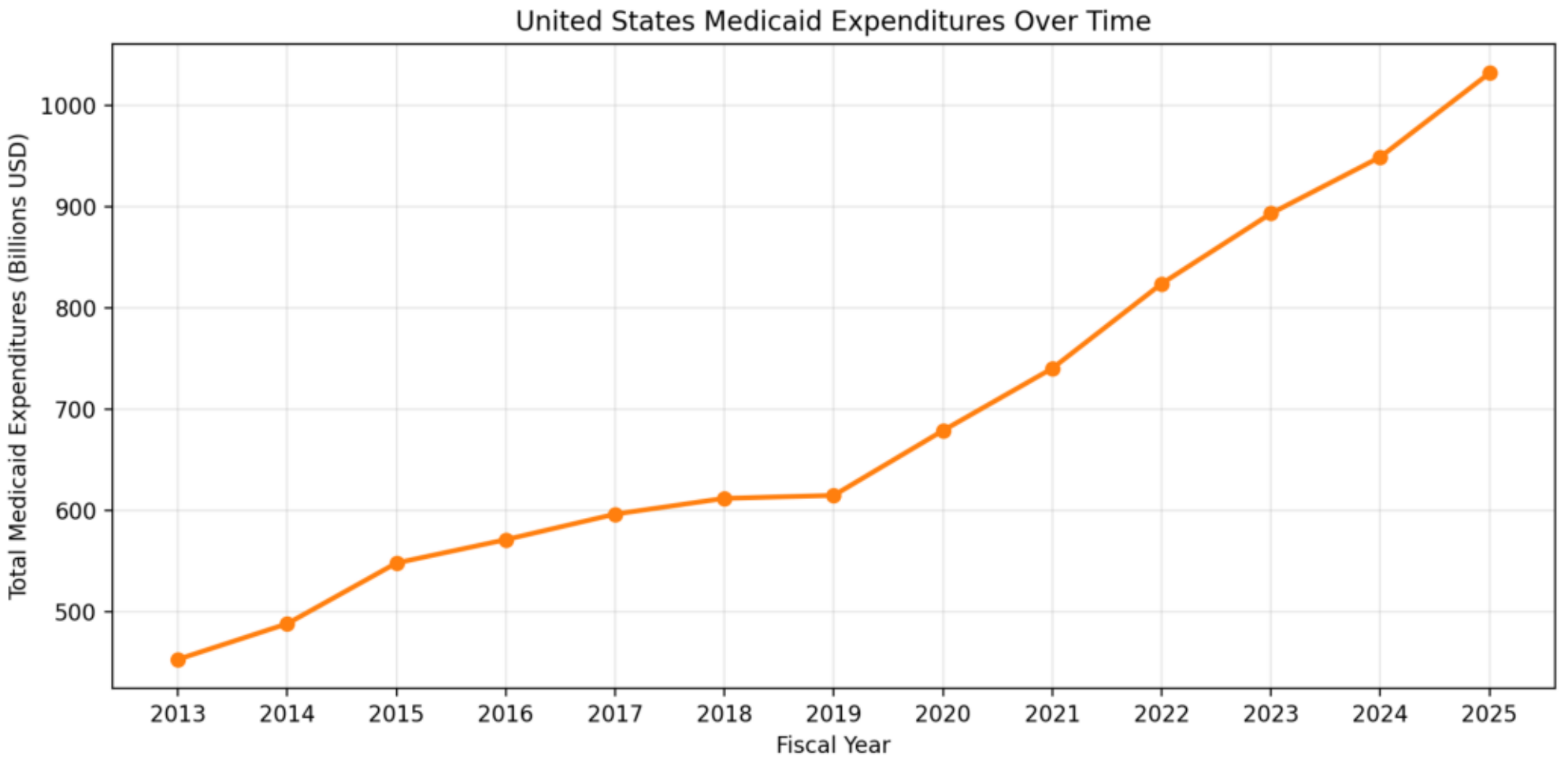
Michigan Expenditure Trend

Observed Michigan Medicaid expenditures across the historical window.



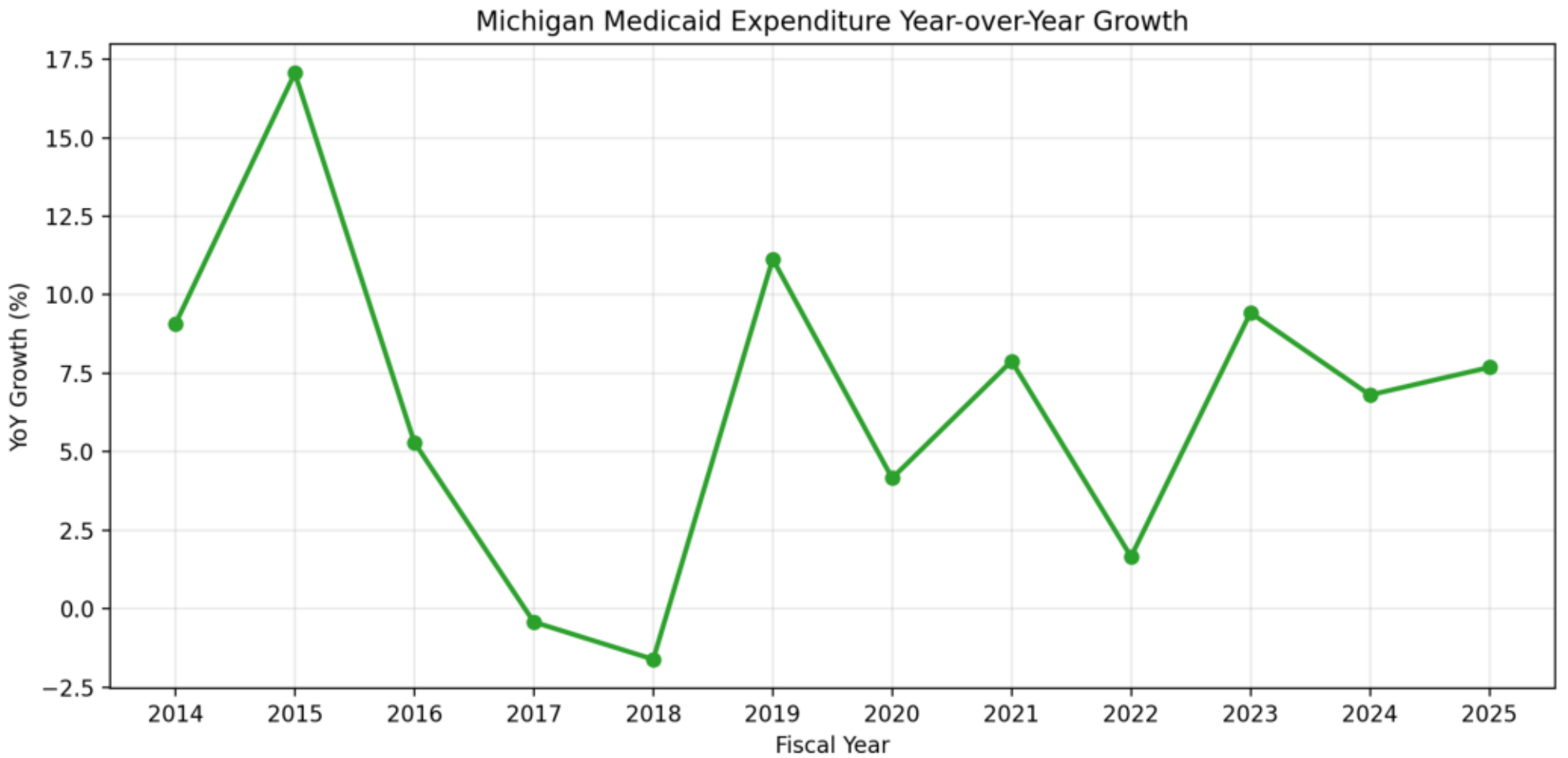
National Expenditure Trend

Observed United States Medicaid expenditures across the historical window.



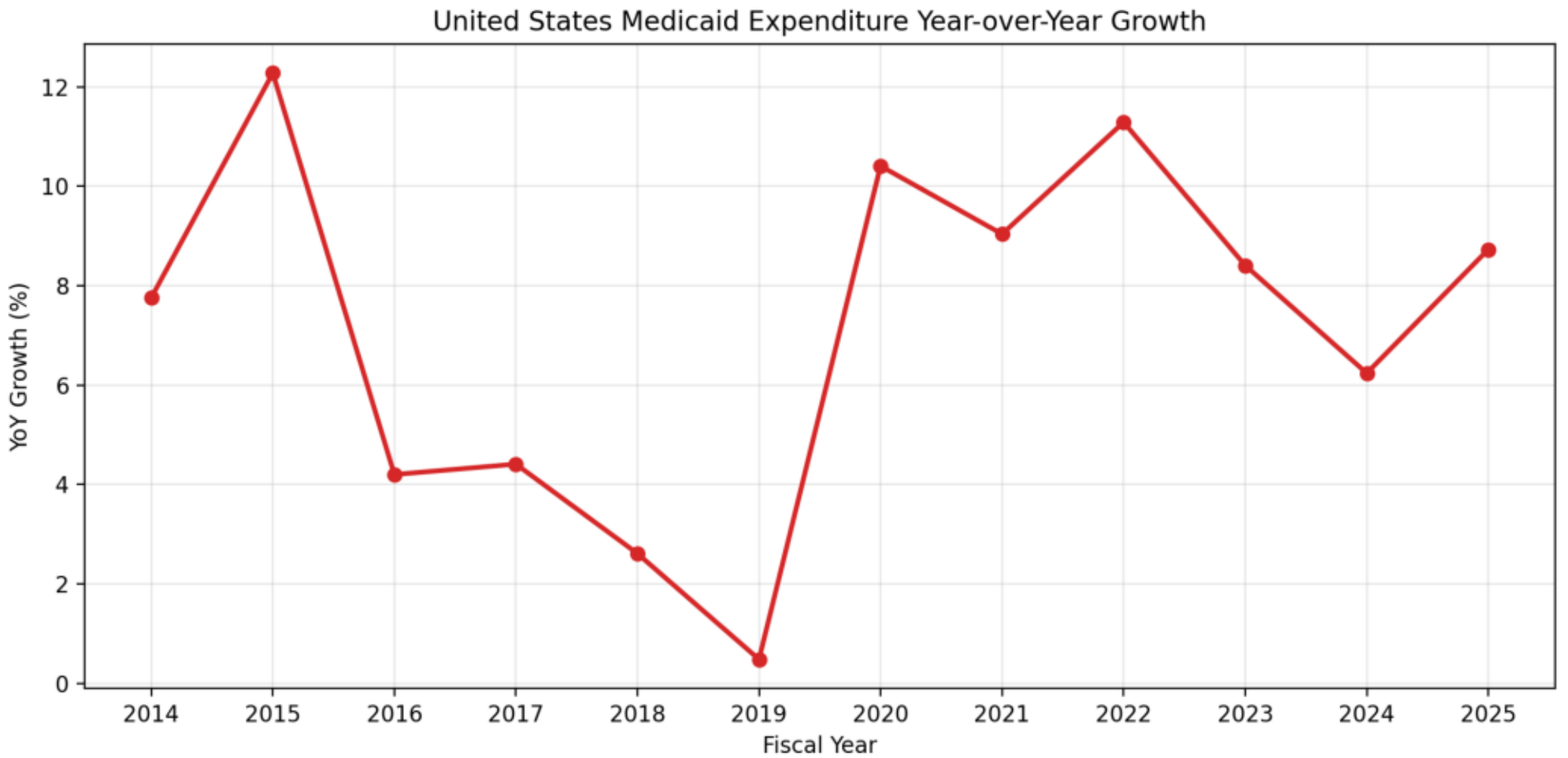
Michigan YoY Growth

Year-over-year growth in Michigan Medicaid expenditures.



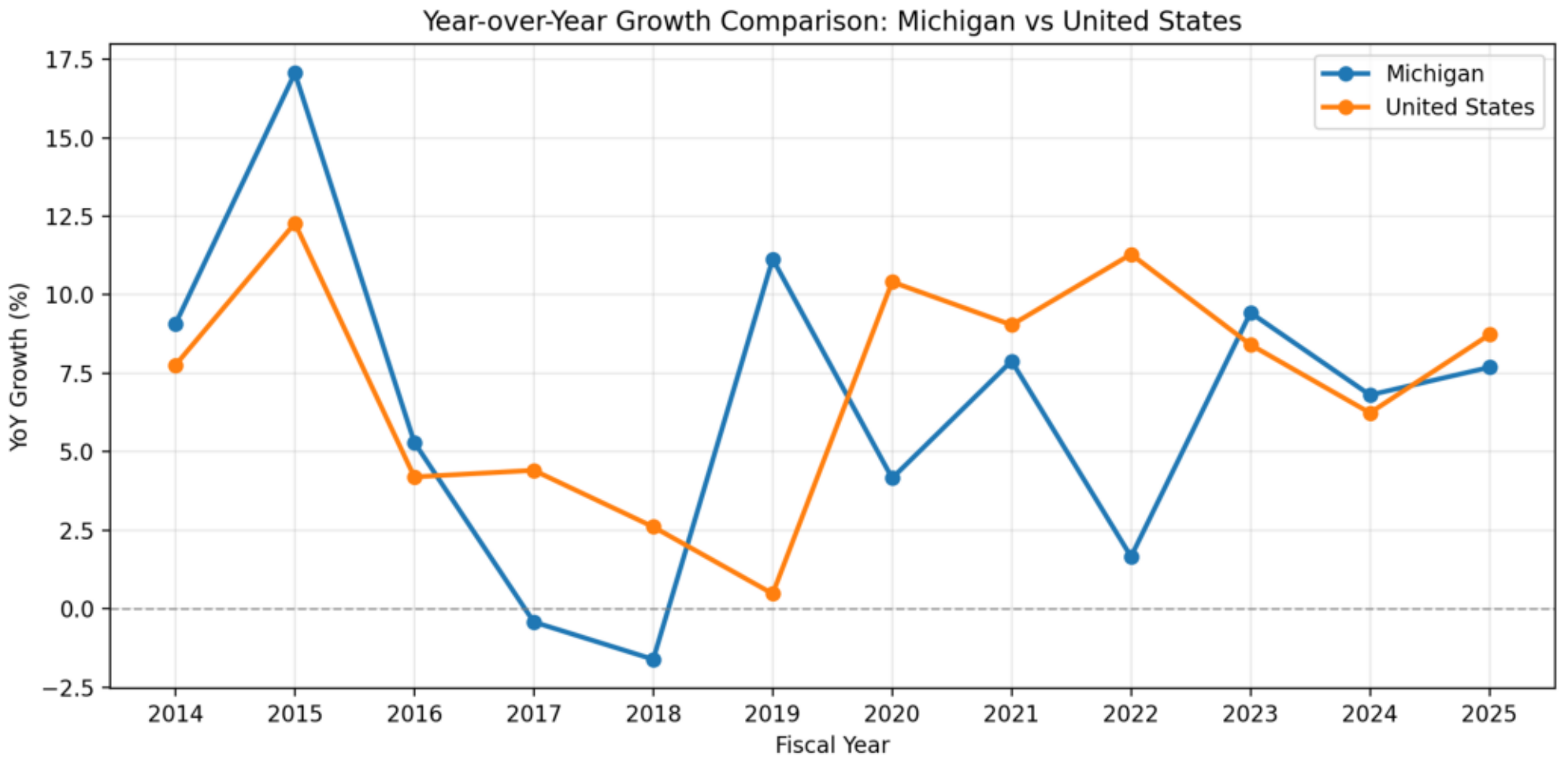
National YoY Growth

Year-over-year growth in United States Medicaid expenditures.



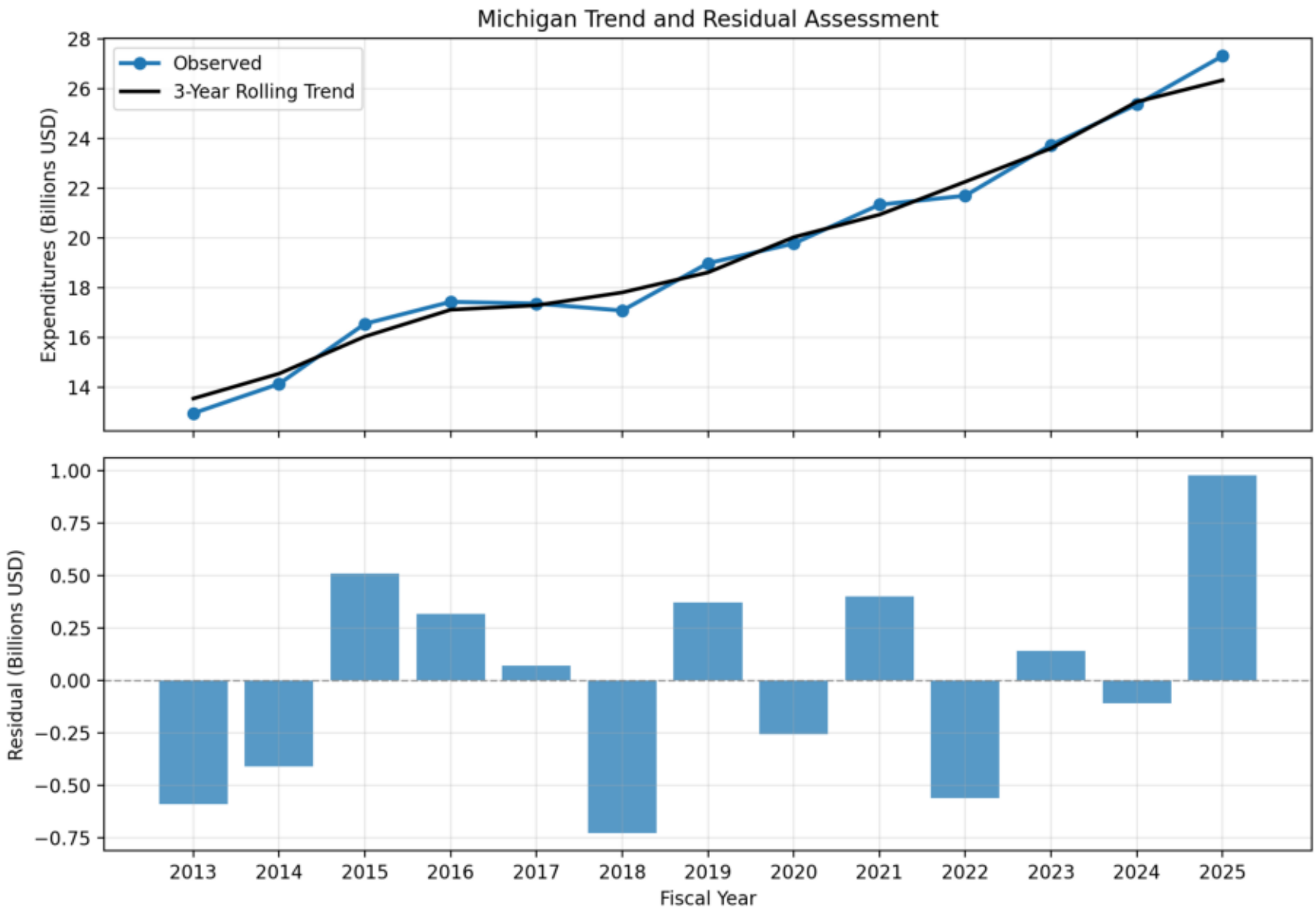
Michigan vs National YoY Growth

Comparison of annual growth rates between Michigan and the United States.



Michigan Trend and Residual Assessment

Trend and residual view used to assess structural movement in annual data.



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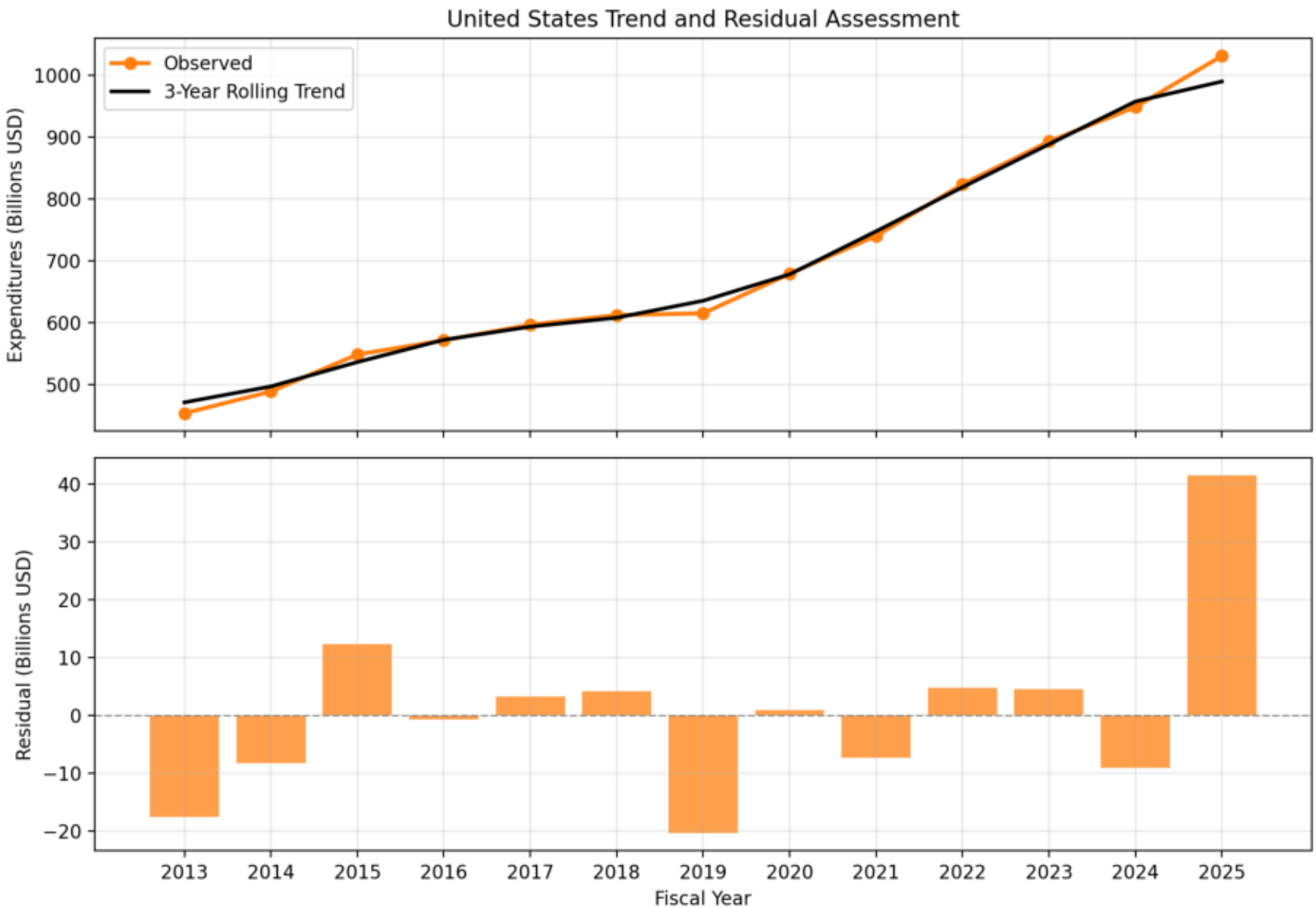

Baseline Forecast Holdout Visualizations

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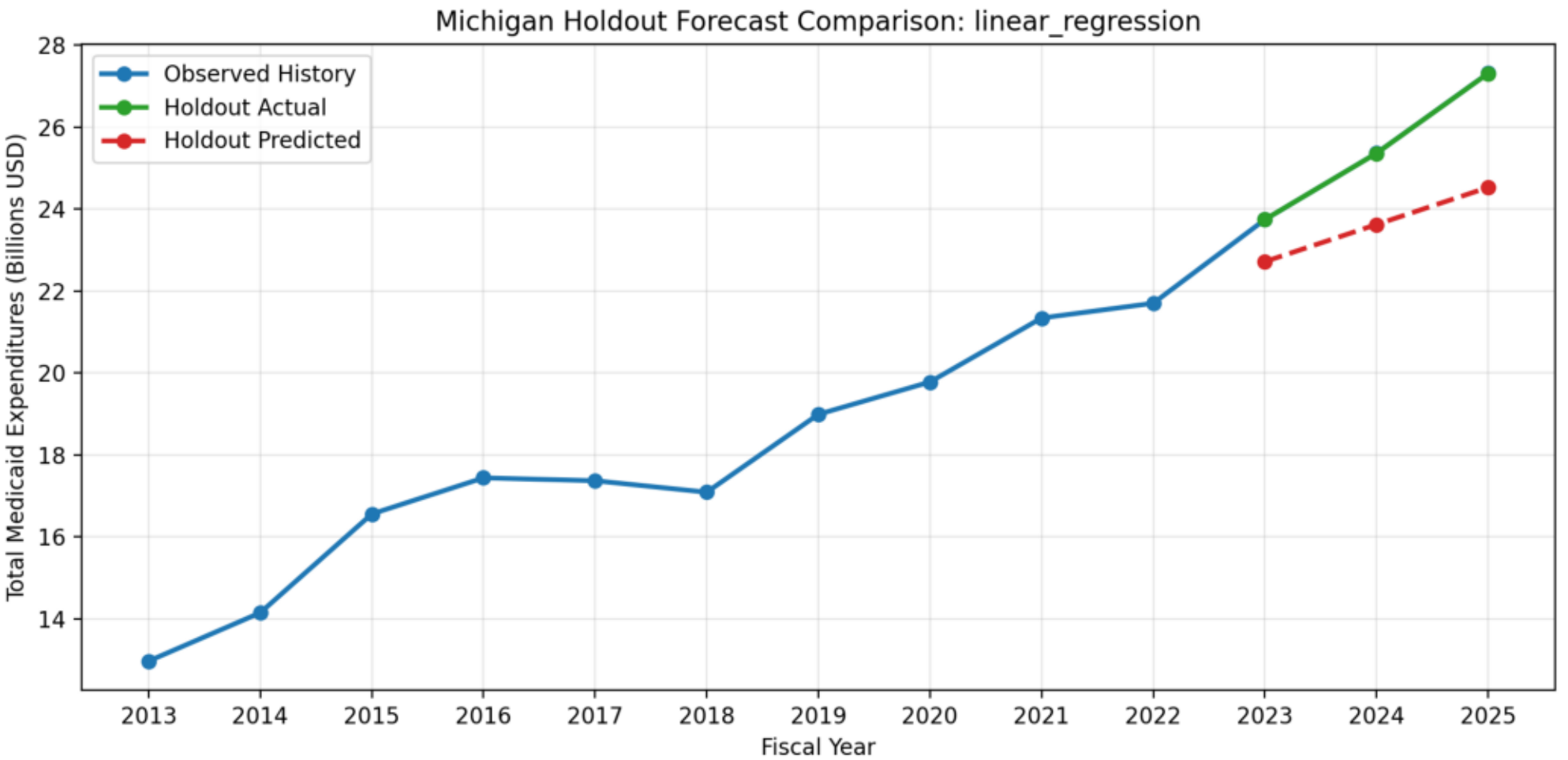
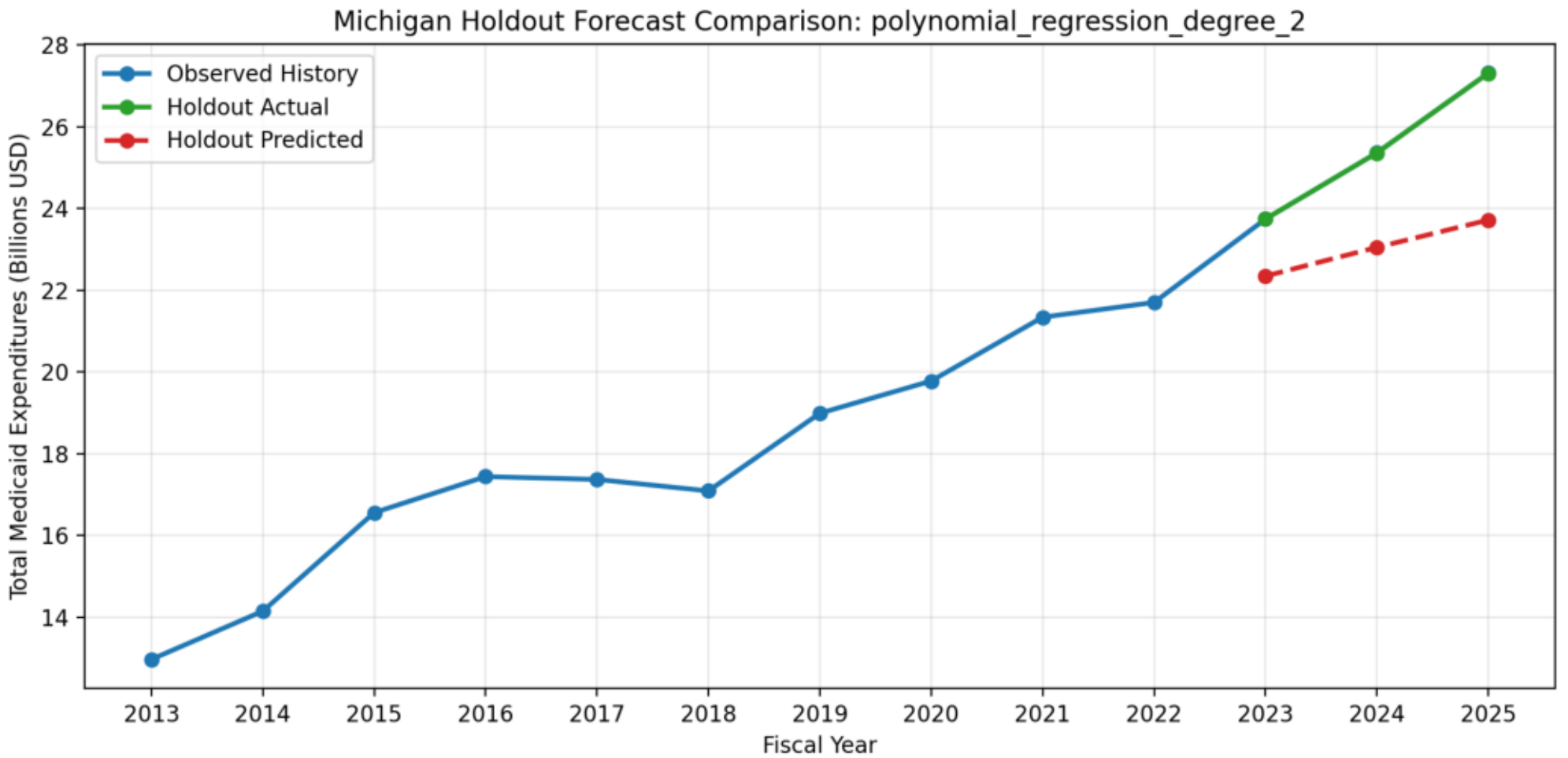
Michigan Linear Regression Holdout

Holdout comparison for the linear regression baseline.



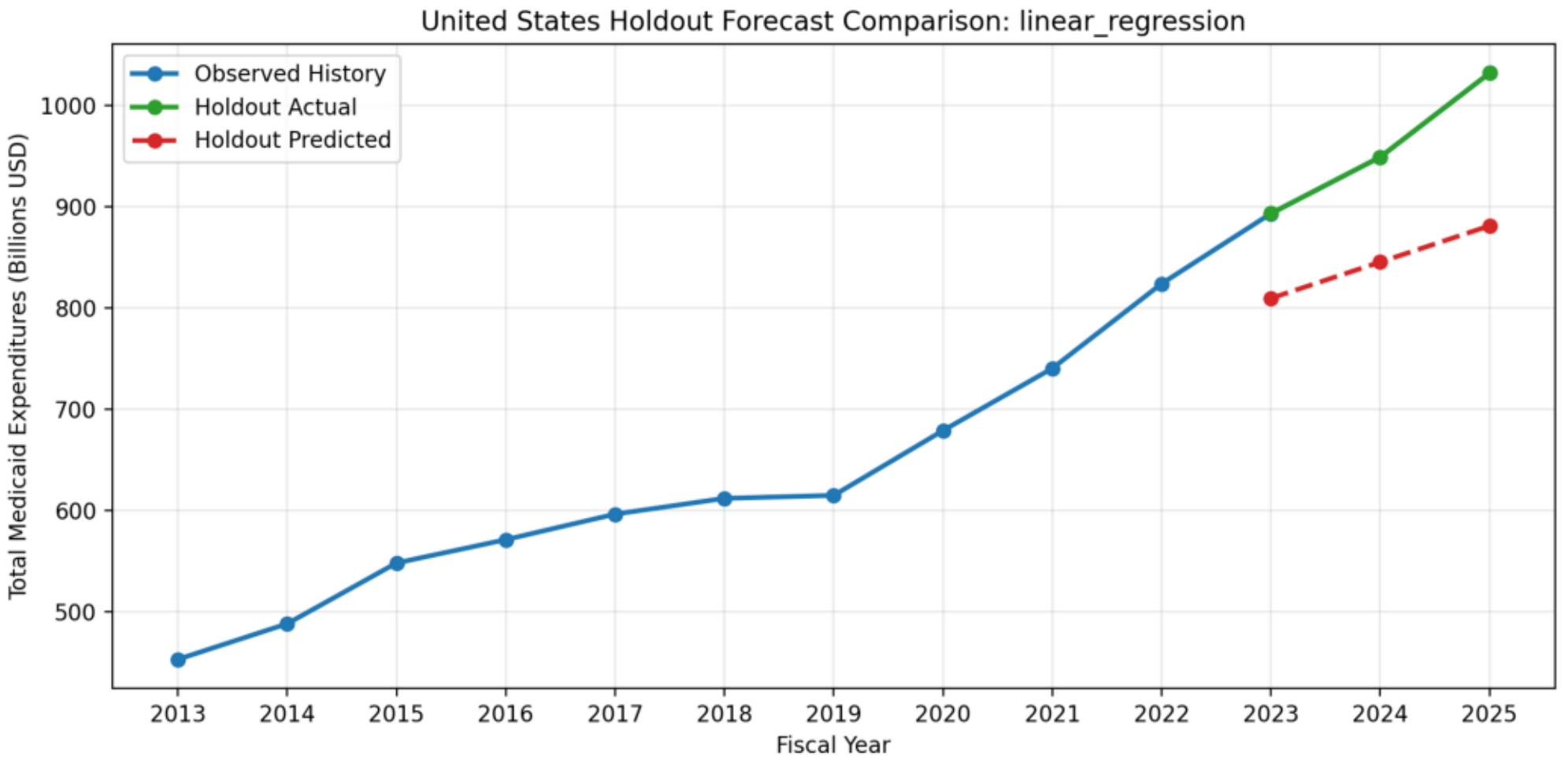
Michigan Polynomial Regression Holdout

Holdout comparison for the second-degree polynomial baseline.



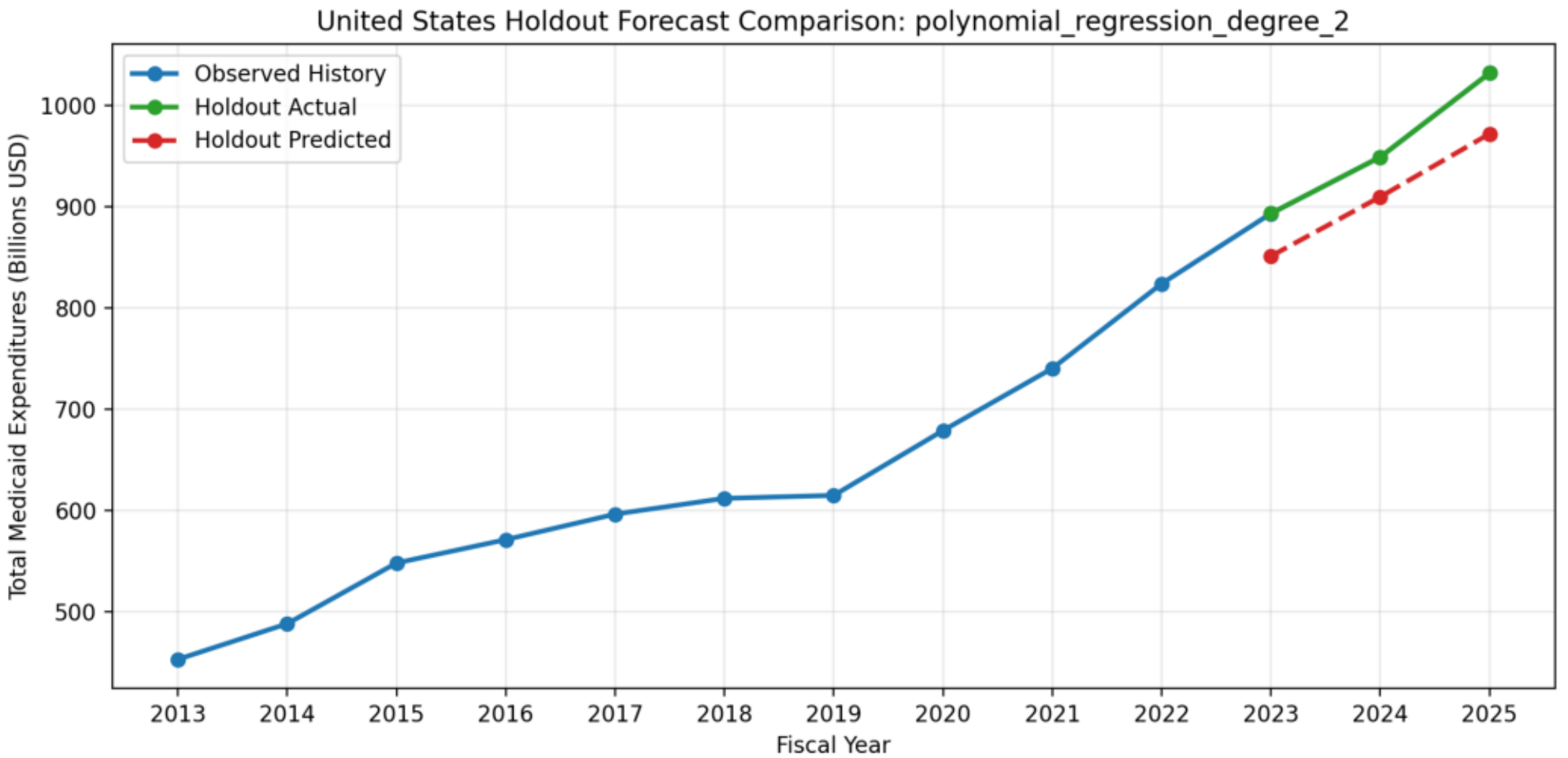
United States Linear Regression Holdout

Holdout comparison for the linear regression baseline.



United States Polynomial Regression Holdout

Holdout comparison for the second-degree polynomial baseline.



Time-Series Forecast Holdout Visualizations

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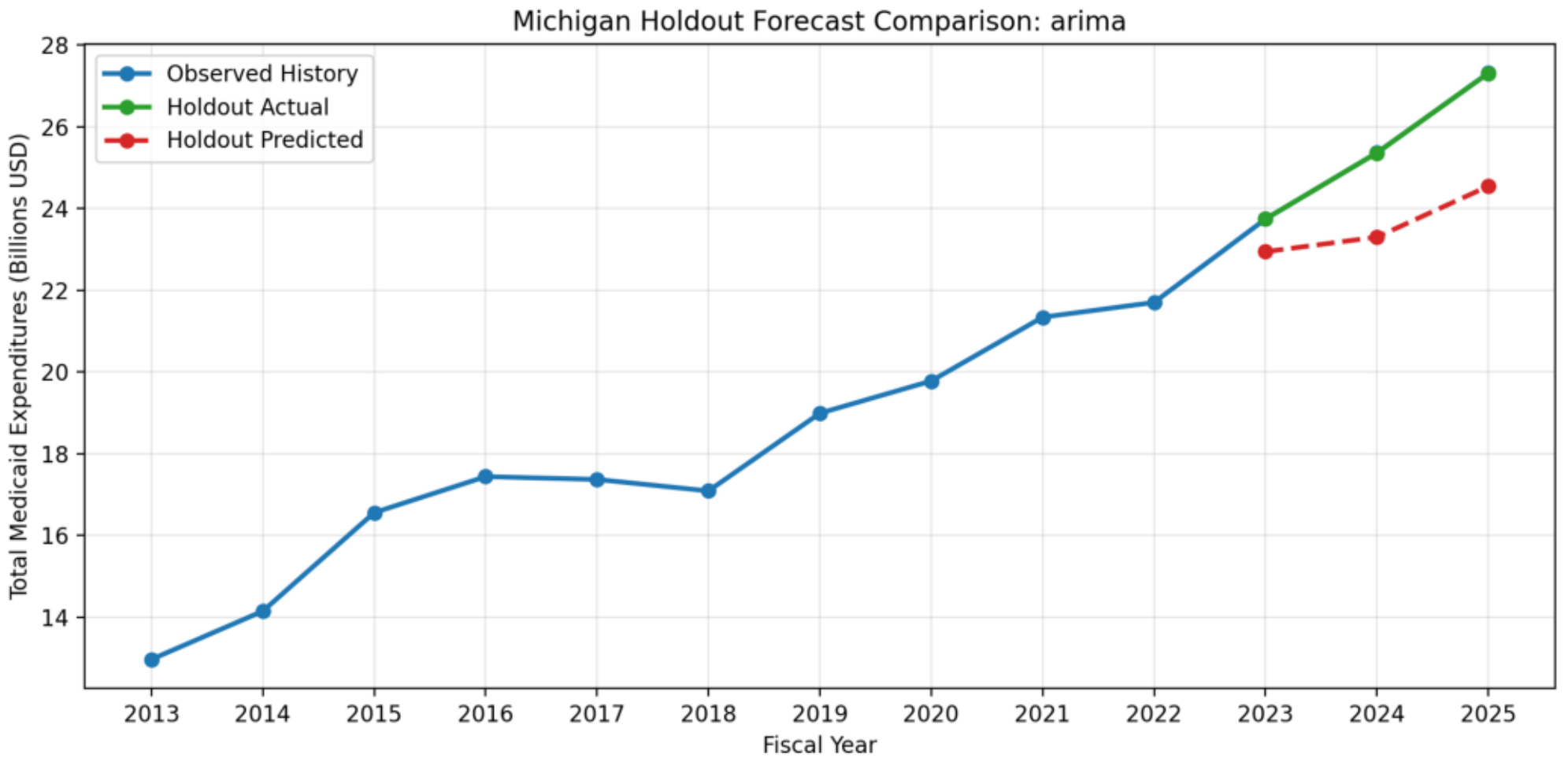
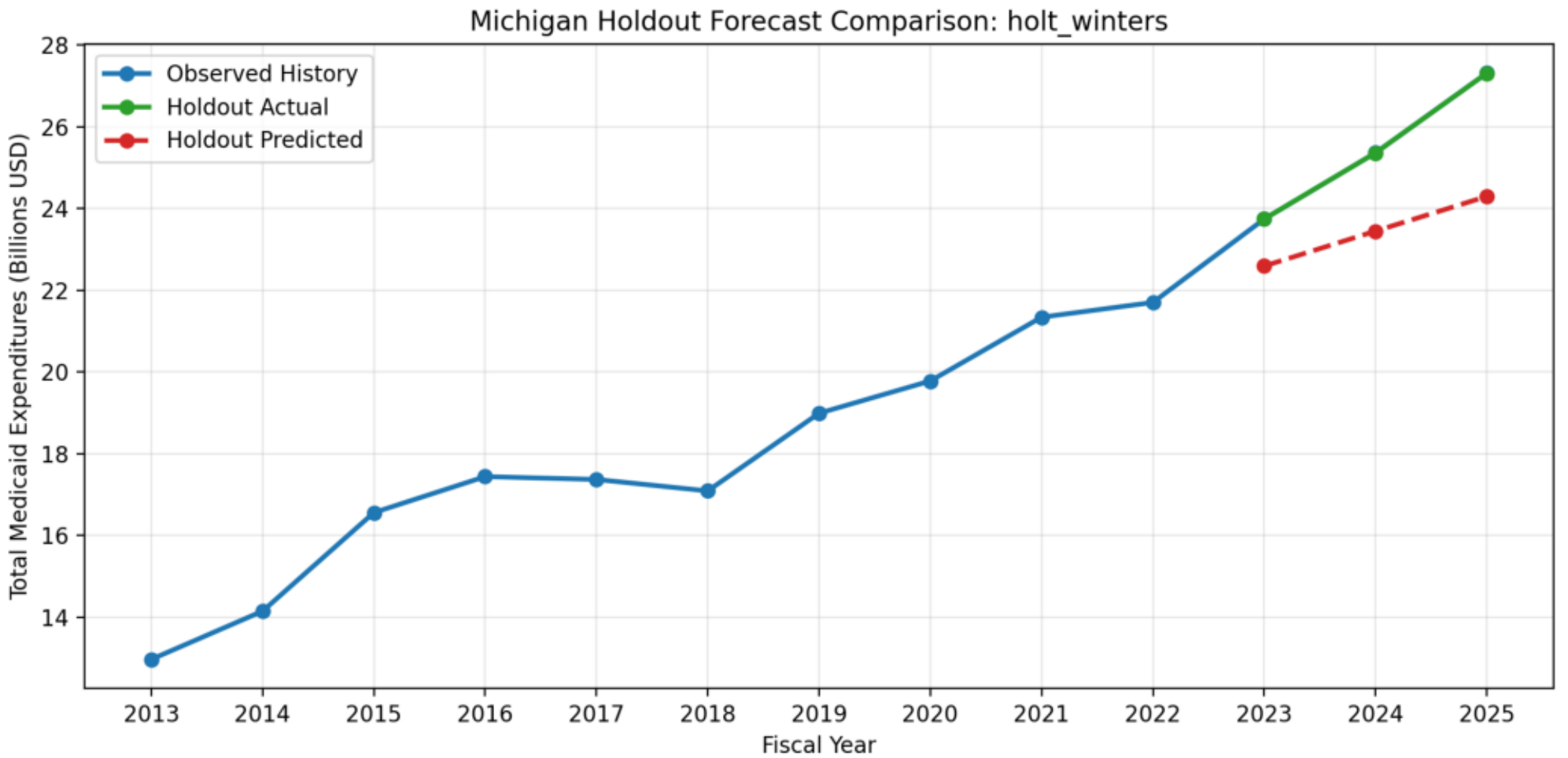
Michigan ARIMA Holdout

Holdout comparison for the ARIMA model.



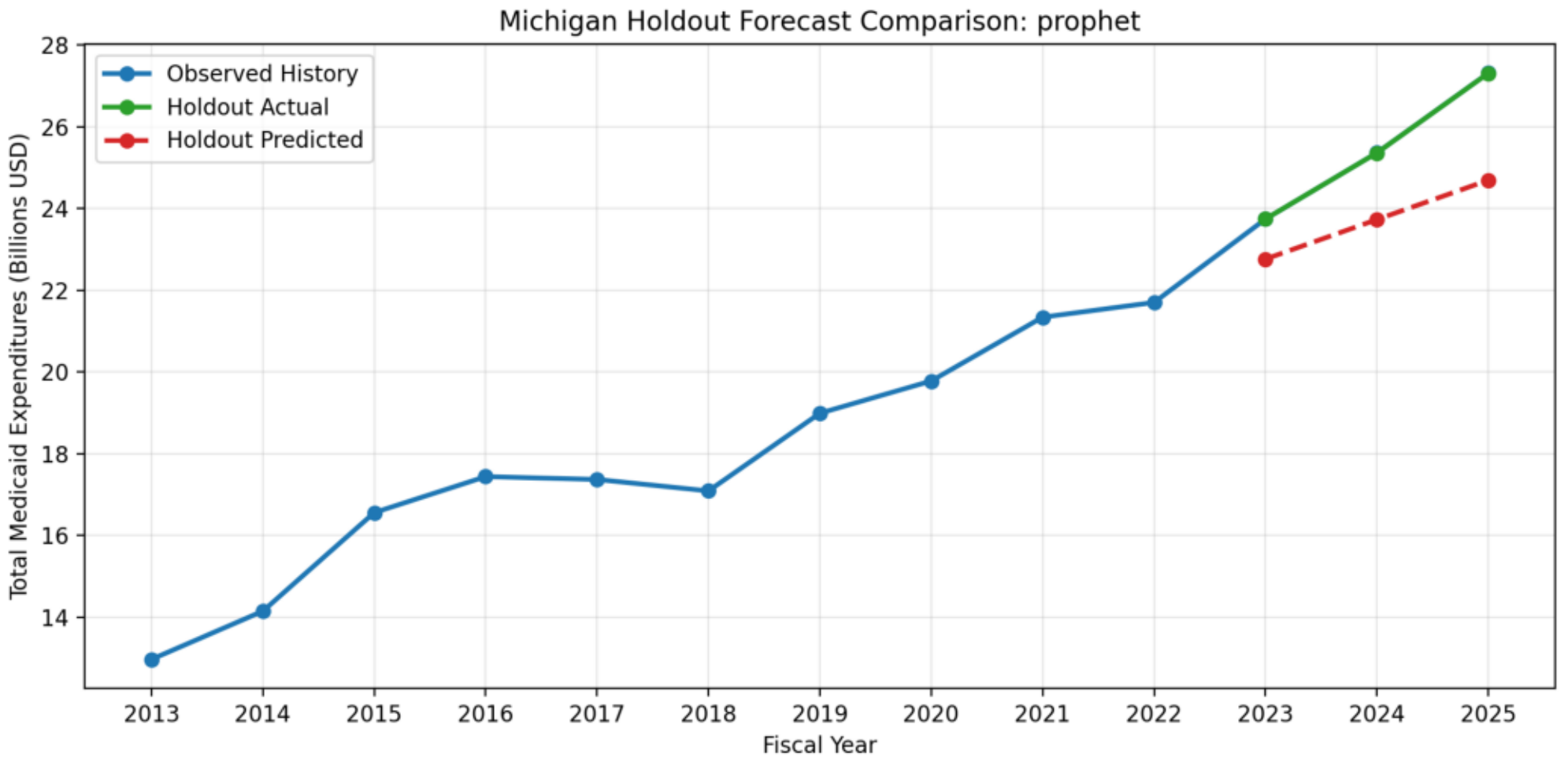
Michigan Holt-Winters Holdout

Holdout comparison for the Holt-Winters model.



Michigan Prophet Holdout

Holdout comparison for the Prophet model.



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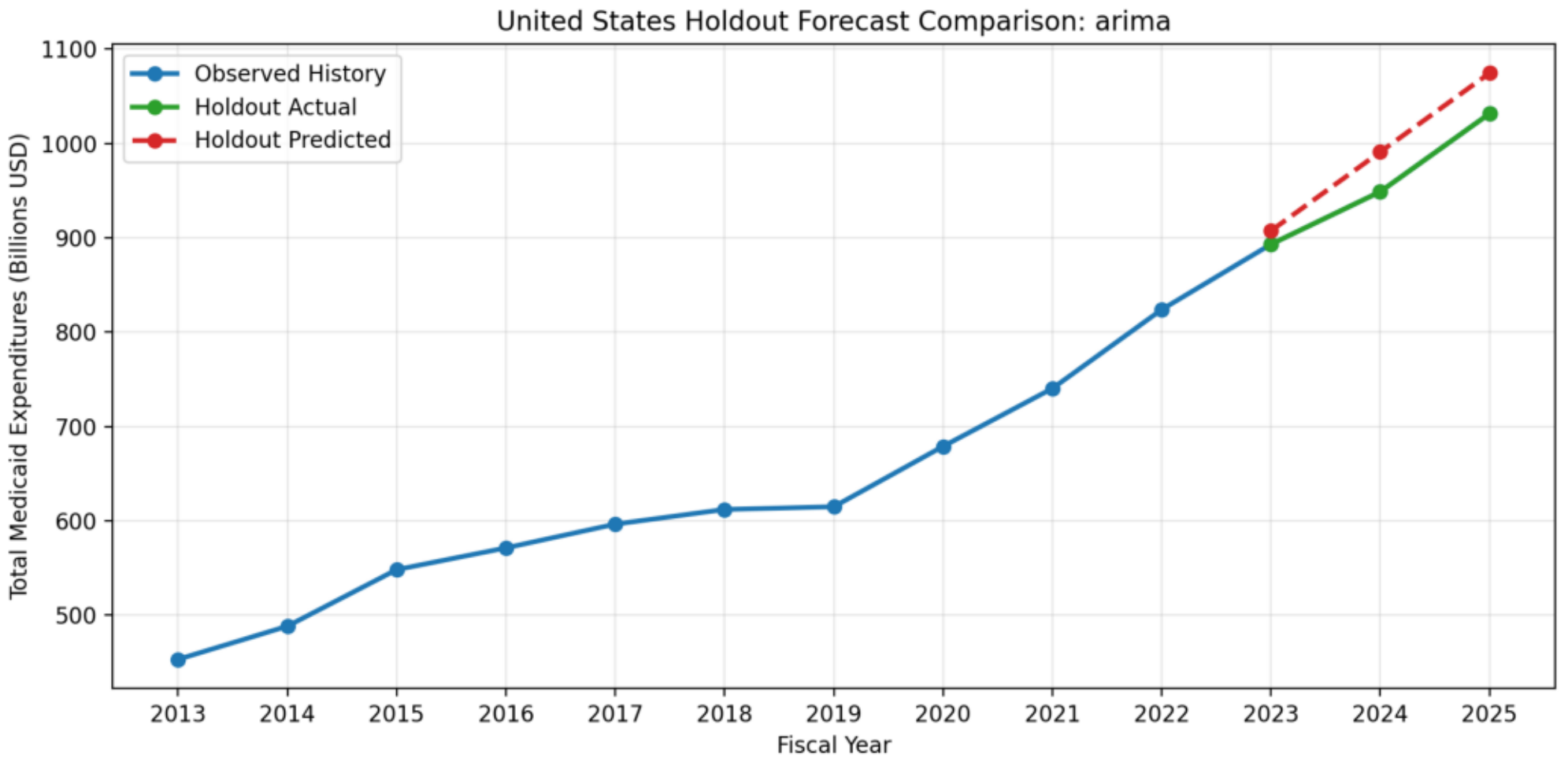

United States Holt-Winters Holdout

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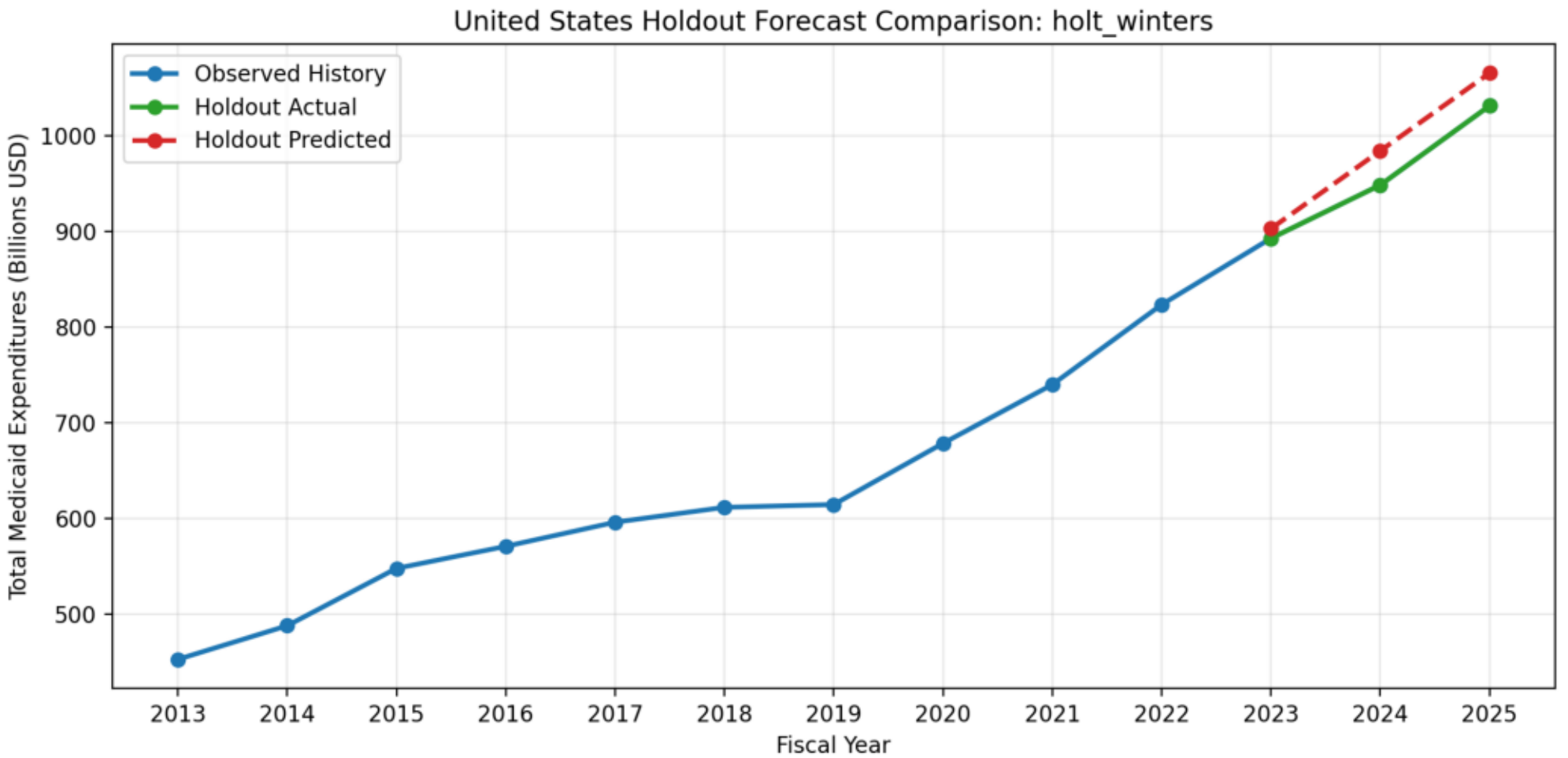
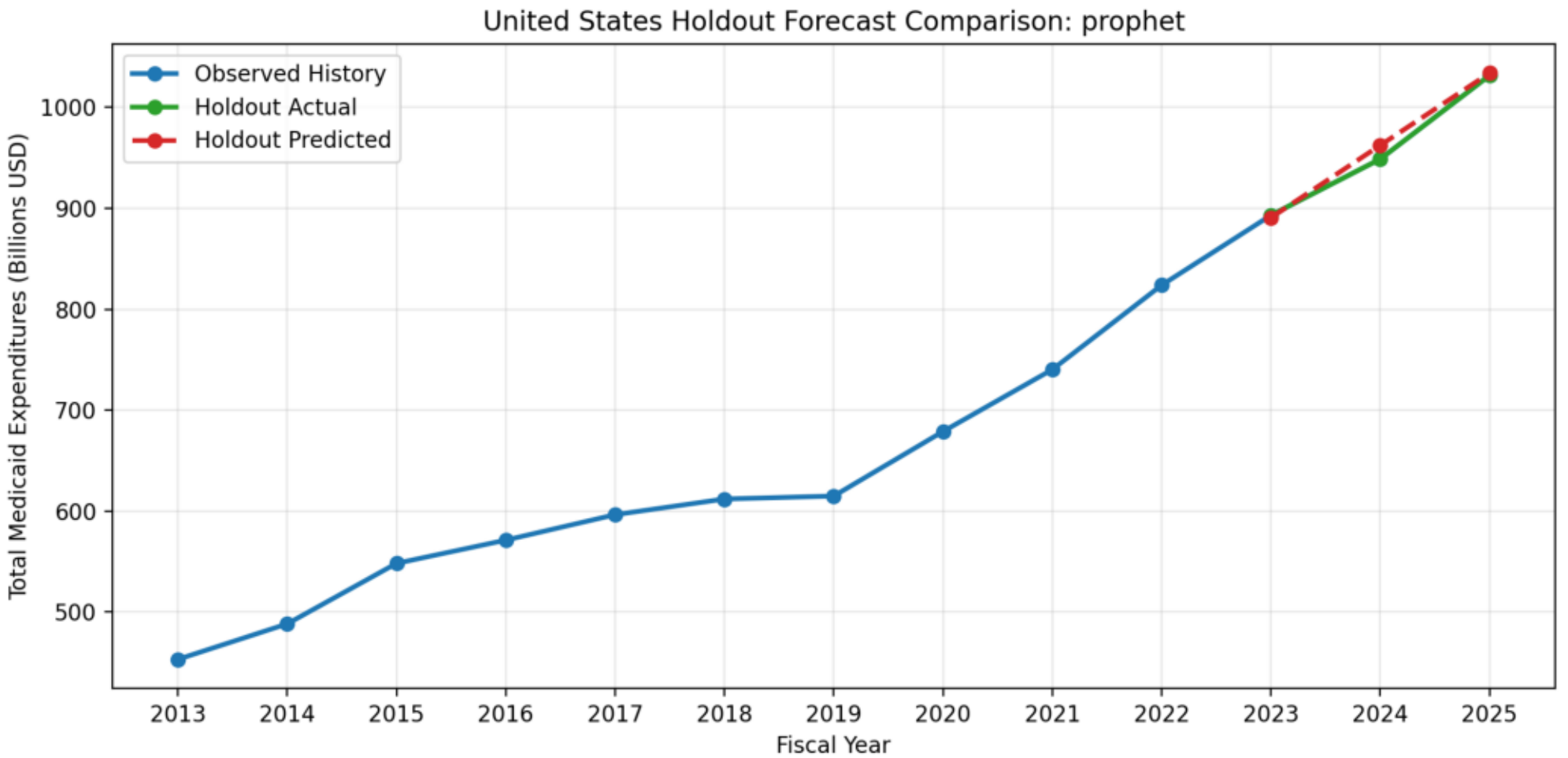


United States Prophet Holdout

Holdout comparison for the Prophet model.



Model Selection Comparison Charts

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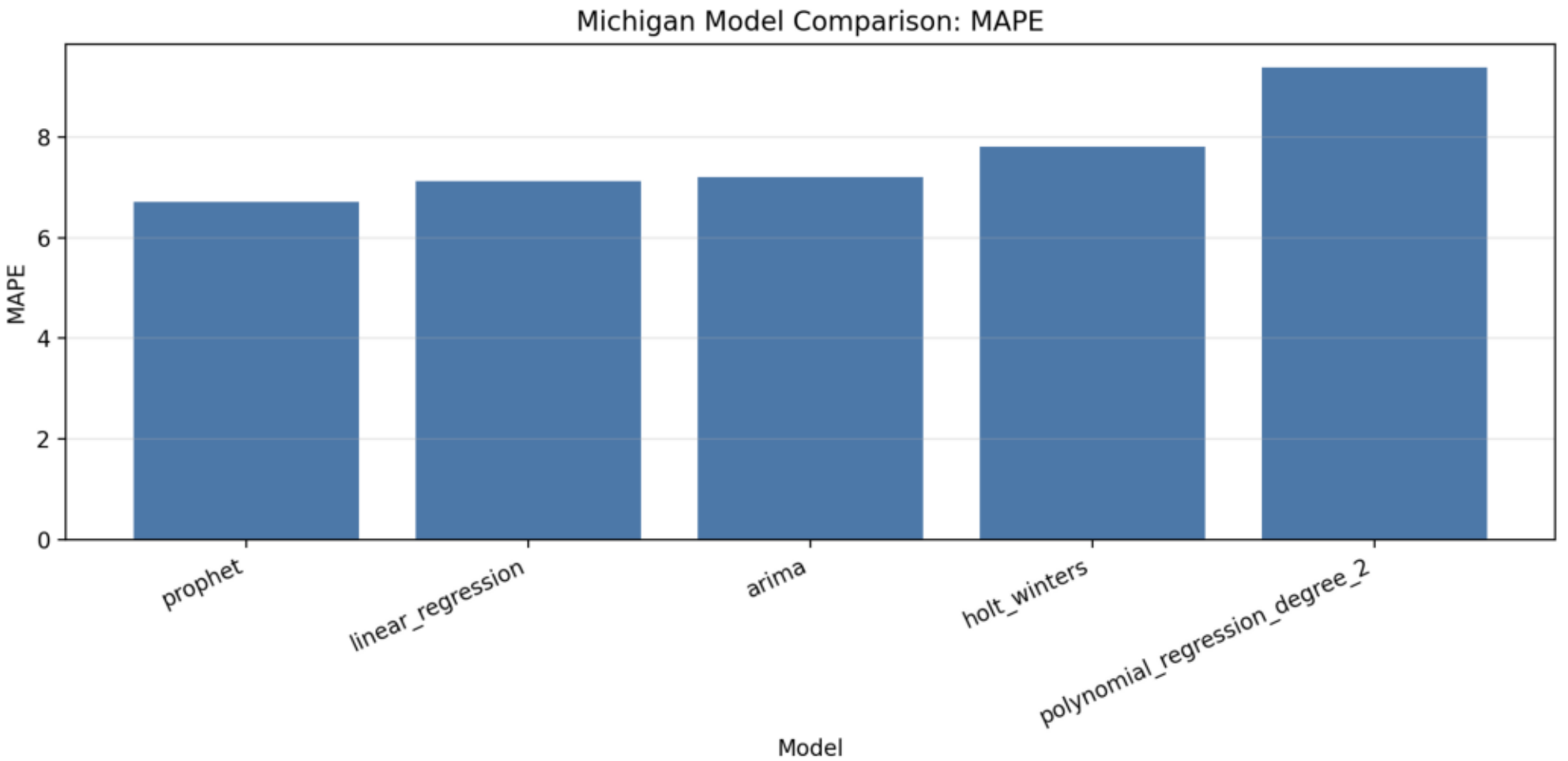
Michigan MAPE Comparison

Lower MAPE indicates better holdout performance.



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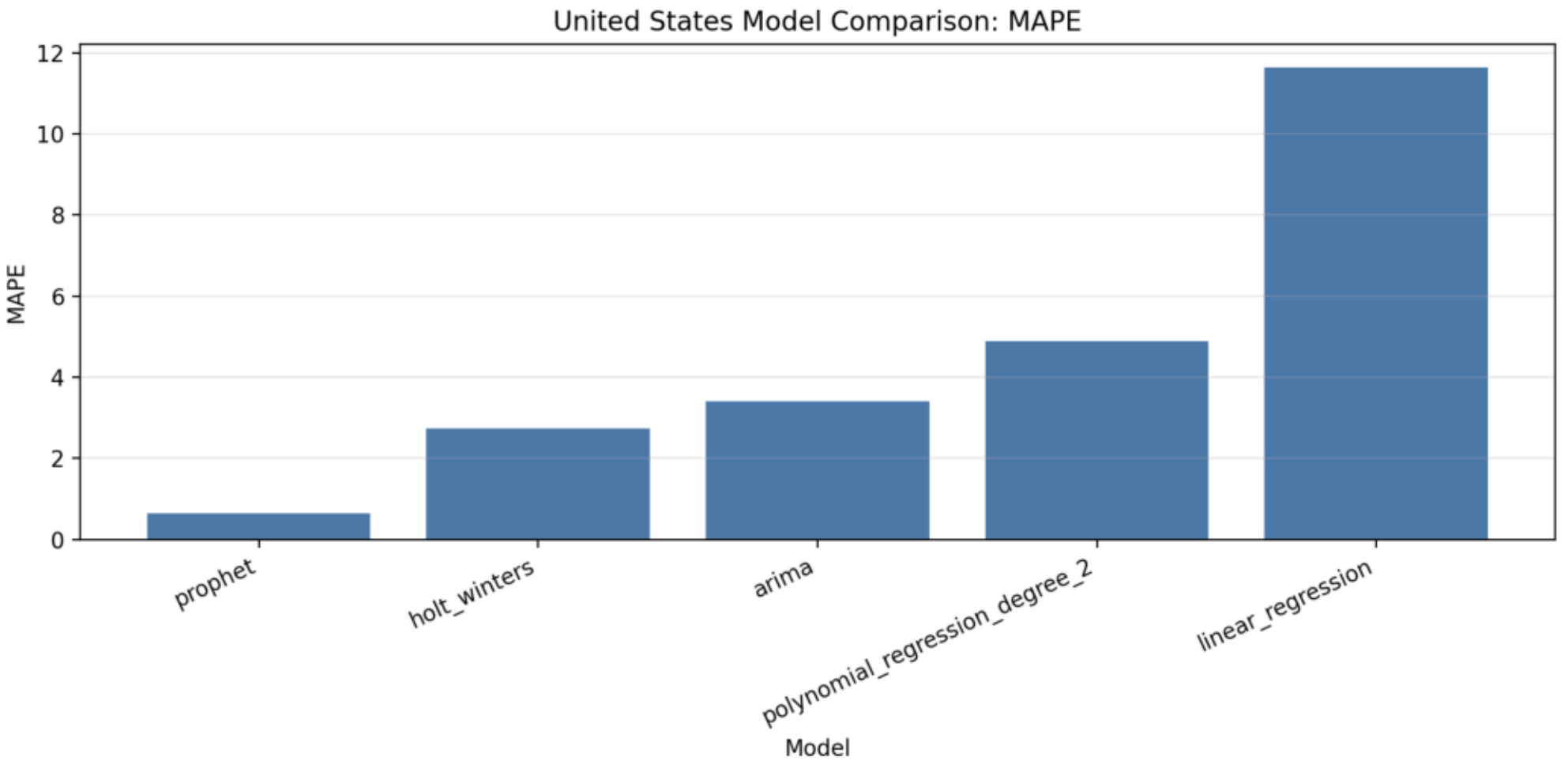
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